3.6 TRACING OF PARAMETRIC CURVES

The points to be taken into consideration while tracing a parametric curve $x = f_1(t)$, $y = f_2(t)$, where t is a parameter are as follows:

(i) Symmetry

(a) The curve is symmetric about x-axis if x is an even function and y is an odd function of t.

(b) The curve is symmetric about y-axis if x is an odd function and y is an even function of t.

(c) The curve is symmetric about y-axis if after replacing t by $\pi - t$, x becomes negative and y remains positive.

(ii) Origin

The curve passes through the origin if there exists at least one real value of t at which $x = 0$ and $y = 0$.

(iii) Points of Intersection

(a) Points of intersection with x-axis: Find the value of t at which $y = 0$ and then find x for this value of t.

(b) Points of intersection with y-axis: Find the value of t at which $x = 0$ and then find y for this value of t.

(iv) Tangents

(a) Tangent is parallel to x-axis at the point where $\dfrac{dy}{dx} = 0$.

(b) Tangent is parallel to y-axis at the point where $\dfrac{dy}{dx} \to \infty$.

(v) Maximum and Minimum Values

Determine the maximum and minimum values of x and y if exists.

(vi) Region

Determine the region where x and y are real. The curve does not exist in the region, where x or y is imaginary.

(vii) Variation of x and y

Determine the values of x and y for some suitable values of t.

Note: If x and y are periodic functions of t having the same period, then the curve is traced for one period only.

Example 1

Trace the hyplocycloid $x = a\cos^3 t$, $y = b\sin^3 t$.

Solution

x and y are periodic functions of t with period 2π. Hence, the curve is traced between 0 to 2π.

(i) *Symmetry:* The curve is symmetric about x-axis since x is an even function of t and y is an odd function of t. Also the curve is symmetric about y-axis since after replacing t by $\pi - t$, x becomes negative but y remains positive.

(ii) *Origin:* The curve does not pass through the origin.

(iii) *Points of Intersection:*

(a) At $t = 0$, $y = 0$ and $x = a$.

(b) At $t = \frac{\pi}{2}$, $x = 0$ and $y = b$.

Thus, the curve meets the x-axis at $A(a, 0)$ and y-axis at $B(0, b)$.

(iv) *Tangents:* $\frac{dx}{dt} = -3a\cos^2 t \sin t$, $\frac{dy}{dt} = 3b\sin^2 t \cos t$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3b\sin^2 t\cos t}{-3a\cos^2 t\sin t} = -\frac{b}{a}\tan t$$

$\frac{dy}{dx} = 0$, when $t = 0$

Thus, the tangent is x-axis at $t = 0$ i.e., at $A(a, 0)$.

$\frac{dy}{dx} \to \infty$ when $t = \frac{\pi}{2}$.

Thus, the tangent is y-axis at $t = \frac{\pi}{2}$, i.e., at $B(0, b)$.

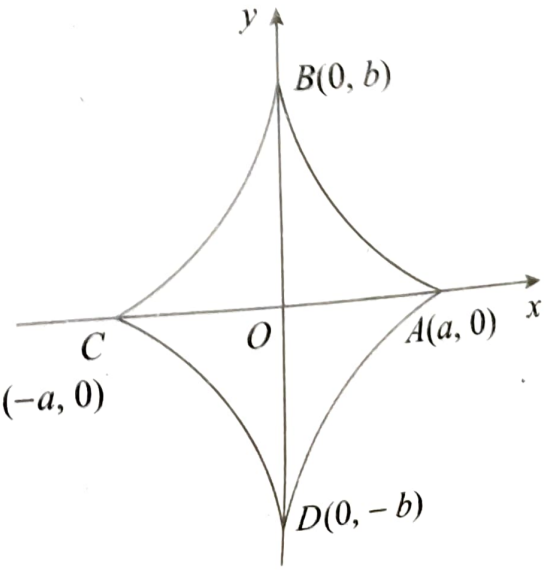
(v) *Maximum and Minimum Values:* Maximum values of x and y are a and b respectively since maximum value of $\cos t$ and $\sin t$ is 1. Minimum values of x and y are $-a$ and $-b$ respectively since minimum value of $\cos t$ and $\sin t$ is -1.

(vi) *Region:* The curve lies in the region $-a < x < a$ and $-b < y < b$.

(vii) *Asymptotes:* There is no asymptote of the curve since x and y are finite for all values of t.

(viii) *Variation of x and y:*

t	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
x	a	$\frac{3\sqrt{3}\,a}{8}$	$\frac{a}{2\sqrt{2}}$	$\frac{a}{8}$	0
y	0	$\frac{b}{8}$	$\frac{b}{2\sqrt{2}}$	$\frac{3\sqrt{3}\,b}{8}$	b



Example 2

Trace the tractrix $x = a\left[\cos t + \log\left|\tan\left(\frac{t}{2}\right)\right|\right]$, $y = a\sin t$.

Solution

(i) *Symmetry:* The curve is symmetric about x-axis since x is an even function of t and y is an odd function of t.

Replacing t by $\pi - t$,

$$x = a\left[\cos(\pi - t) + \log\left|\tan\left(\frac{\pi}{2} - \frac{t}{2}\right)\right|\right]$$

$$= a\left[-\cos t + \log\left|\cot\frac{t}{2}\right|\right]$$

$$= a\left[-\cos t - \log\left|\tan\frac{t}{2}\right|\right]$$

$$= -x$$

$$y = a\sin(\pi - t)$$

$$= a\sin t$$

$$= y$$

Thus, the curve is symmetric about y-axis.

(ii) *Origin:* The curve does not pass through the origin.

(iii) *Points of Intersection:*

(a) At $t = 0$, $y = 0$ and $x \to -\infty$ [$\because \log 0 \to -\infty$]

(b) At $t = \frac{\pi}{2}$, $x = 0$ and $y = a$

Thus, the curve meets the y-axis at $A(0, a)$ and does not meet y-axis.

(iv) *Tangents:*

$$x = a\left[\cos t + \frac{1}{2}\log\tan^2\frac{t}{2}\right]$$

$$\frac{dx}{dt} = a\left[-\sin t + \frac{1}{2}\cdot\frac{1}{\tan^2\frac{t}{2}}\cdot 2\tan\frac{t}{2}\sec^2\frac{t}{2}\cdot\frac{1}{2}\right]$$

$$= a\left[-\sin t + \frac{1}{\sin t}\right] = \frac{a\cos^2 t}{\sin t}$$

$$\frac{dy}{dt} = a\cos t$$

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$= \frac{a\cos t\cdot\sin t}{a\cos^2 t} = \tan t$$

At point $A(0, a)$: $\dfrac{dy}{dx} = \tan\dfrac{\pi}{2} \to \infty$

Thus, the tangent is y-axis.

(v) *Maximum and Minimum Values:* Maximum and minimum values of y are a and $-a$ respectively since maximum and minimum values of $\sin t$ are 1 and -1 respectively.

x lies between $-\infty$ to ∞.

(vi) *Region:* The curve lies in the region $-a < y < a$ and $-\infty < x < \infty$.

(vii) *Asymptotes:* $\lim\limits_{t\to 0} x = -\infty$ and $\lim\limits_{t\to\pi} x = \infty$

At $t = 0$ and $t = \pi$, $y = 0$

Thus, $y = 0$ i.e., x-axis is the asymptote of the curve.

(viii) *Variation of x and y:*

t	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
x	$-\infty$	$-0.45a$	$-0.17a$	$-0.04a$	0
y	0	$0.5a$	$0.71a$	$0.87a$	a

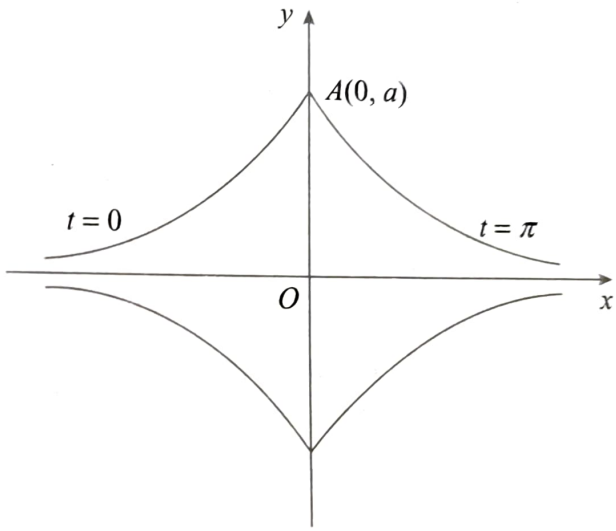


Fig. 3.20

Example 3

Trace the cycloid $x = a(t + \sin t)$, $y = a(1 - \cos t)$.

Solution

(i) *Symmetry:* The curve is symmetric about y-axis since x is an odd function of t and y is and even function of t.

(ii) *Origin:* At $t = 0$, $x = 0$ and $y = 0$. Thus, the curve passes through the origin.

(iii) *Points of Intersection:*

(a) $x = 0$ only at $t = 0$

Thus, the curve meets the y-axis only at origin.

(b) If $y = 0$, $\cos t = 1$, $t = 0, \pm 2\pi, \pm 4\pi$,
Then $x = 0, \pm 2a\pi, \pm 4a\pi, \ldots$
Thus, the curve meets the x-axis at $(0, 0)$, $(\pm 2a\pi, 0)$, $(\pm 4a\pi, 0)$ ...

(iv) *Tangents:*

$$\frac{dx}{dt} = a(1+\cos t), \quad \frac{dy}{dt} = a\sin t$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$= \frac{a\sin t}{a(1+\cos t)} = \frac{2\sin\frac{t}{2}\cos\frac{t}{2}}{2\cos^2\frac{t}{2}} = \tan\frac{t}{2}$$

(a) $\frac{dy}{dx} = 0$, at $t = 0, \pm 2\pi, \pm 4\pi, \ldots$

Thus, tangent is x-axis at $(0, 0)$, $(\pm 2a\pi, 0)$, $(\pm 4a\pi, 0)$, ...

(b) $\frac{dy}{dx} \to \infty$, at $t = \pm\pi, \pm 3\pi, \pm 5\pi, \ldots$

Thus, tangent is parallel to y-axis at $(\pm a\pi, 2a)$, $(\pm 3a\pi, 2a)$, $(\pm 5a\pi, 2a)$, ...

(v) *Maximum and Minimum Values:* Maximum and minimum values of y are $2a$ and 0 since minimum and maximum values of $\cos t$ are -1 and 1.

(vi) *Region*: Curve lies in the region $0 < y < 2a$ and $-\infty < x < \infty$.

(vii) *Asymptotes*: There is no asymptote of the curve since x and y are finite for all finite values of t.

(viii) Since $\sin t$ is a periodic function of period 2π, y is the periodic function of period 2π. Thus, curve repeat itself in the intervals $[0, \pm 2a\pi]$, $[\pm 2a\pi, \pm 4a\pi]$, ...

(ix) *Variation of x and y:*

t	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
x	0	$a\left(\frac{\pi}{2}+1\right)$	$a\pi$	$a\left(\frac{3\pi}{2}+1\right)$	$2a\pi$
y	0	a	$2a$	a	0

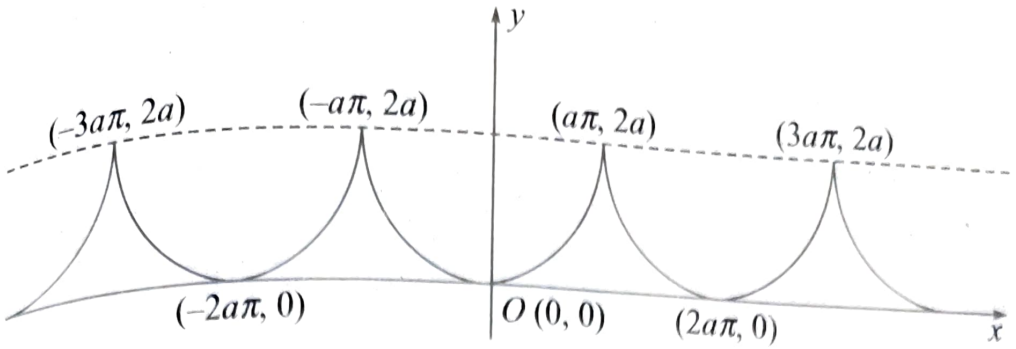


Fig. 3.21

Example 4

Trace the curve $x = a\sin 2t(1 + \cos 2t)$, $y = a\cos 2t(1 - \cos 2t)$.

Solution

x and y are periodic functions of t with period π, hence, we will discuss the curve only in the interval $0 \le t < \pi$

(i) *Symmetry*: Curve is symmetric about y-axis since x is an odd function of t and y is an even function of t.

(ii) *Origin*: At $t = 0$, $x = 0$ and $y = 0$. Thus, origin lies on the curve.

(iii) *Points of Intersection*: (a) $x = 0$ at $t = 0, \pm\frac{\pi}{2}$, then $y = 0, -2a$.

Thus, the curve meets the y-axis at $(0, 0)$, $(0, -2a)$.

(b) $y = 0$ at $t = 0, \pm\frac{\pi}{4}$

then $x = 0, \pm a$

Thus, the curve meets the x-axis at $(0, 0)$, $(a, 0)$, $(-a, 0)$.

(iv) *Tangents*: $\frac{dx}{dt} = 2a\cos 2t(1 + \cos 2t) + a\sin 2t(-2\sin 2t)$

$= 2a\cos 2t + 2a\cos 4t$

$= 2a \cdot 2\cos 3t\cos t$

$\frac{dy}{dt} = -2a\sin 2t(1 - \cos 2t) + a\cos 2t(2\sin 2t)$

$= 4a\sin 2t\cos 2t - 2a\sin 2t$

$= 2a(\sin 4t - \sin 2t)$

$= 2a \cdot 2\cos 3t\sin t$

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

$= \frac{4a\cos 3t\sin t}{4a\cos 3t\cos t}$

$= \tan t$

(a) $\frac{dy}{dx} = 0$, at $t = 0$

Thus, the tangent is x-axis at $(0, 0)$.

(b) $\frac{dy}{dx} \to \infty$ at $t = \frac{\pi}{2}$

Thus, the tangent is y-axis at $(0, -2a)$.

(v) *Asymptotes*: There is no asymptote of the curve since x and y are finite for all values of t.

(vi) *Variation of x and y*

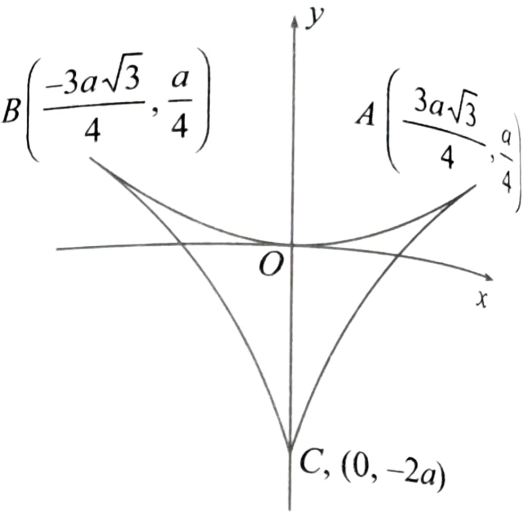


Fig. 3.22

t	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$	π
x	0	$\frac{3a\sqrt{3}}{4}$	$\frac{a\sqrt{3}}{4}$	0	$\frac{-a\sqrt{3}}{4}$	$\frac{-3\sqrt{3}}{4}$	0
y	0	$\frac{a}{4}$	$\frac{-3a}{4}$	$-2a$	$\frac{-3a}{4}$	$\frac{a}{4}$	0

EXERCISE 3.2

Trace the following curves:

1. Astroid $x = a\cos^3 t$, $y = b\sin^3 t$

Ans.:

